

```

#include <iostream>

using namespace std;

void read(int sensors[], int n) {
    cout << "Enter 8 sensor readings: ";

    for (int i = 0; i < n; i++) {
        cin >> sensors[i];
    }
}

float pos(int sensors[], int n) {
    int sum = 0;
    int count = 0;

    for (int i = 0; i < n; i++) {
        if (sensors[i] == 1) {
            sum += i;
            count++;
        }
    }

    if (count == 0)
        return -1;

    return (float)sum / count;
}

void Movement(float position) {
    if (position == -1) {
        cout << "Line Position: Lost" << endl;
        cout << "Action: Line Lost" << endl;
    }

    else if (position < 3.5) {

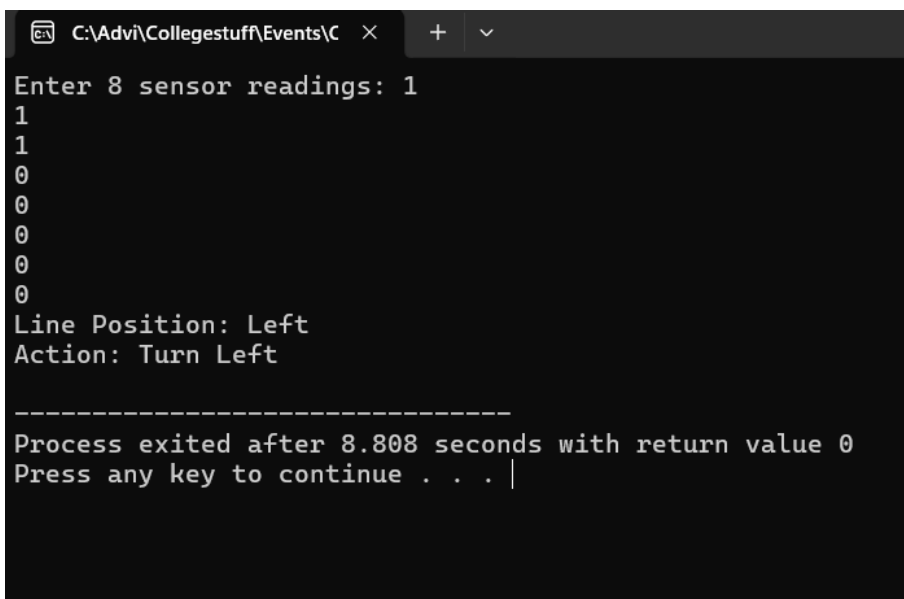
```

```

        cout << "Line Position: Left" << endl;
        cout << "Action: Turn Left" << endl;
    }
    else if (position > 3.5) {
        cout << "Line Position: Right" << endl;
        cout << "Action: Turn Right" << endl;
    }
    else {
        cout << "Line Position: Center" << endl;
        cout << "Action: Move Forward" << endl;
    }
}

int main() {
    int sensors[8];
    read(sensors, 8);
    float position = pos(sensors, 8);
    Movement(position);
    return 0;
}

```



```

C:\Advi\Collegestuff\Events\C >
Enter 8 sensor readings: 1
1
1
0
0
0
0
0
0
Line Position: Left
Action: Turn Left

-----
Process exited after 8.808 seconds with return value 0
Press any key to continue . . . |

```

```
#include <iostream>

using namespace std;

class Robot {

private:

    int speed;

    int battery;

public:

    void setSpeed(int s) {
        if (s >= 0 && s <= 100)
            speed = s;
        else
            speed = 0;
    }

    void setBattery(int b) {
        if (b >= 0 && b <= 100)
            battery = b;
        else
            battery = 0;
    }

    void moveForward() {
        if (battery == 0) {
            cout << "Battery Empty!" << endl;
            return;
        }
        cout << "Moving Forward" << endl;
        battery -= 5;
    }
}
```

```
void moveBackward() {  
    if (battery == 0) {  
        cout << "Battery Empty!" << endl;  
        return;  
    }  
    cout << "Moving Backward" << endl;  
    battery -= 5;  
}  
  
void turnLeft() {  
    if (battery == 0) {  
        cout << "Battery Empty!" << endl;  
        return;  
    }  
    cout << "Turning Left" << endl;  
    battery -= 5;  
}  
  
void turnRight() {  
    if (battery == 0) {  
        cout << "Battery Empty!" << endl;  
        return;  
    }  
    cout << "Turning Right" << endl;  
    battery -= 5;  
}  
  
void displayStatus() {  
    cout << "Robot Speed: " << speed << endl;  
    cout << "Battery: " << battery << "%" << endl;  
}
```

```
};
```

```
int main() {  
    Robot r;  
    int speed, battery;  
    cout << "Enter Speed: ";  
    cin >> speed;  
    cout << "Enter Battery: ";  
    cin >> battery;  
    r.setSpeed(speed);  
    r.setBattery(battery);  
    r.moveForward();  
    r.turnLeft();  
    r.moveForward();  
    r.turnRight();  
    r.displayStatus();  
    return 0;  
}
```

```
Enter Speed: 50  
Enter Battery: 99  
Moving Forward  
Turning Left  
Moving Forward  
Turning Right  
Robot Speed: 50  
Battery: 79%
```

```
-----  
Process exited after 7.519 seconds with return value 0  
Press any key to continue . . . |
```